Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 102605 | *Subject: Automation in Manufacturing* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) What is the main objective of modeling and simulation in manufacturing?
(i) To optimize production processes
(ii) To increase labour cost
(iii) To eliminate product design
(iv) To optimize the design

(b) Which aspect of CNC technology must directly enables adaptive control in manufacturing processes?
(i) Mechanical structure
(ii) Manual programming
(iii) Integrated feedback systems
(iv) Batch processing

(c) In flexible automation, which component is crucial for managing changes in product designs efficiently?
(i) Rigid fixtures
(ii) CAD software
(iii) Assembly lines
(iv) Hydraulic Systems

(d) What is a distinct advantage of implementing computer graphics software in CAD over traditional design methods?
(i) Reduced tool wear
(ii) Lower energy consumption
(iii) Fast Data transfer
(iv) Rapid prototyping

(e) Which optimization technique is frequently applied in simulation modeling for manufacturing process improvement?
(i) Adaptive control logic
(ii) Process scheduling
(iii) Linear regression
(iv) Computer-integrated manufacturing (CIM)

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(f) Which of the following is an example of low-cost automation?
(i) CNC machining center
(ii) Pneumatic system
(iii) Computer-aided design
(iv) Flexible manufacturing system

(g) What is the primary function of PLC in manufacturing automation?
(i) To design products
(ii) To control and monitor industrial processes
(iii) To simulate product routes
(iv) To handle material manually

(h) What is the main advantage of flexible automation over rigid automation?
(i) Higher initial cost
(ii) Requires more human intervention
(iii) Slower production rate
(iv) Ability to handle product variety

(i) Which technology is used for automated material handling in flexible automation systems?
(i) Conveyor belts
(ii) Robots
(iii) Pneumatic systems
(iv) All of these

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(j) What does process route modelling help determine in product manufacturing?

(i) The tool selection for operations
(ii) The optimal sequence of manufacturing tasks
(iii) The deployment of worker
(iv) The machine sequencing

Q.2 (a) What are different types of automation? Explain the need of automation in industries with suitable examples. [7]

(b) Explain low-cost automation. [7]

Q.3 (a) Discuss the role of CNC and adaptive control in flexible manufacturing systems [7]

(b) Discuss geometric modeling techniques used in CAD for analysis and manufacturing [7]

Q.4 (a) Explain the hardware and software components used in Computer Aided Design (CAD) [7]

(b) Explain pneumatic and hydraulic automation with their usage. [7]

Q.5 (a) Explain CNC technology with its application in industries. [7]

(b) Write short notes on mechanical and electromechanical Systems. [7]

Q.6 (a) Write short notes on CAD, CAM and CIM. [7]

(b) Discuss the working of PLC and microcontroller with its applications. [7]

Q.7 (a) Discuss NC part programming with its advantages. Also write any five G-code and its meaning. [7]

(b) Explain process route modelling. [7]

Q.8 (a) Discuss the modelling and simulation techniques used in process design [7]

(b) Explain the need and current trends of automation in manufacturing [7]

Q.9 (a) What is an optimization technique? Explain any one with example. [7]

(b) Explain automated material handling system. [7]

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